

■ 2.6.4 The Operation of Output Forms

There are two steps in every *Mathematica* calculation. First, *Mathematica* takes your input, and *evaluates* it, performing whatever mathematical operations you have specified. After that, it takes the result, and formats it for output.

Functions like `TeXForm` affect only the output formatting step. If you type in something like `TeXForm[expr]`, the expression `expr` will be evaluated, but the function `TeXForm` will stay wrapped around it, and will not evaluate in any way. However, when it reaches the output formatting step, the presence of the `TeXForm` function serves as a signal to the *Mathematica* output formatter to show the expression in $\text{T}_{\text{E}}\text{X}$ form.

This sets the value of `t` to be an expression enclosed in the function `TeXForm`. When `t` prints out, the presence of `TeXForm` signals the output formatter to generate $\text{T}_{\text{E}}\text{X}$ form.

```
In[1]:= t = TeXForm[a/b]
Out[1]//TeXForm= {a\over b}
```

In full form, you can see the function `TeXForm` as part of the value of `t`.

```
In[2]:= FullForm[t]
Out[2]//FullForm= TeXForm[Times[a, Power[b, -1]]]
```

It is very common to want to see an expression you are working with in various different output forms. Usually, when you apply a function like `TeXForm` to an expression, the function remains “wrapped” around your expression, and to recover your expression, you have to “dig” inside the function. In the particular case when your expression is the current output line `%`, *Mathematica* has a special mechanism which allows you to avoid “digging” inside functions like `TeXForm` to get your expression.

Mathematica recognizes when the outermost function in an output expression is the name of an output form (`InputForm`, `TeXForm`, etc.). In this case, *Mathematica* sets the value of `Out[n]` to be not the whole output expression, but instead the expression inside the output form function. This behavior is reflected in the special output label that *Mathematica* uses in this case. Instead of writing just `Out[n]=` at the beginning of your output, *Mathematica* writes something like `Out[n]//TeXForm=`. This label represents the fact that `TeXForm[Out[n]]` is the expression you have generated. As a result, `Out[n]` is the part *inside* the `TeXForm` function. This means that when you ask for `%`, or when you refer to the output line later on, you get your expression *without* `TeXForm` wrapped around it.

The current output line is set to be the expression `a^2/b^2`, without `FortranForm` wrapped around it.

```
In[3]:= FortranForm[(a/b)^2]
Out[3]//FortranForm= a**2/b**2
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

© 1988 Wolfram Research, Inc. Permission is hereby granted for web users to make one paper copy of this page for their personal use. Further reproduction, or any copying of machine-readable files (including this one) to any server computer, is strictly prohibited.

% refers to the expression, without the `FortranForm` wrapped around it.

`In[4] := %^2`

$$\text{Out}[4] = \frac{a^4}{b^4}$$